

HOW MUSCLES WORK TOGETHER

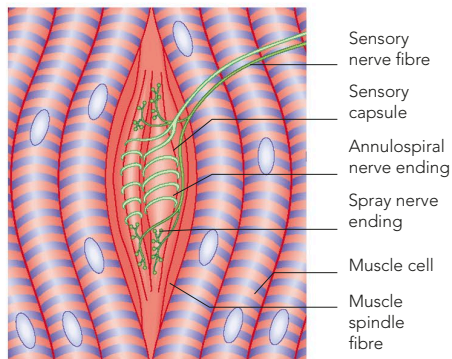
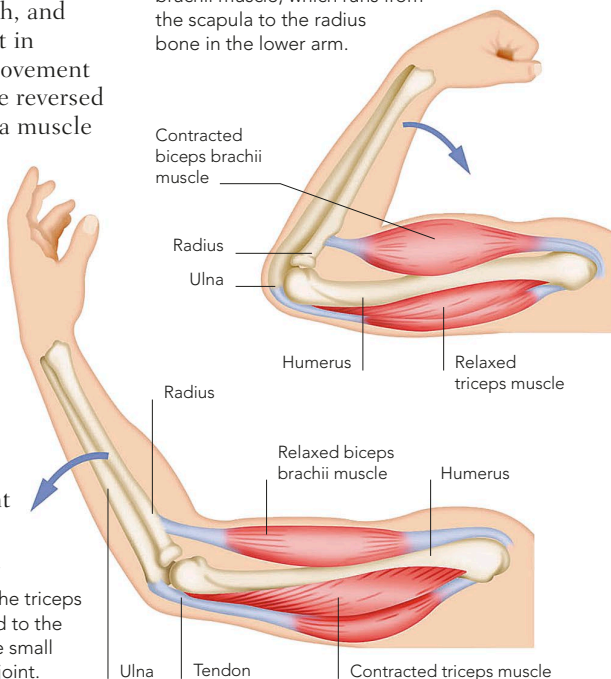
Muscles can only pull, not push, and so are arranged in pairs that act in opposition to one other. The movement produced by one muscle can be reversed by its opposing partner. When a muscle contracts to produce movement, it is called the agonist, while its opposite partner, called the antagonist, relaxes and is passively stretched. In reality, few movements are achieved by a single muscle contraction. Usually, whole teams of muscles act as agonists to give the precisely required degree and direction of motion, while the antagonists tense to prevent the movement over-extending.

STRAIGHTENING THE ELBOW

The biceps brachii relaxes and the triceps brachii, attached at its lower end to the ulna, contracts. It is aided by the small anconeus muscle on the elbow joint.

BENDING THE ELBOW

The chief agonist is the biceps brachii muscle, which runs from the scapula to the radius bone in the lower arm.



NEUROMUSCULAR SPINDLE

These stretch sensors lie between and in parallel with skeletal muscle fibres; information from them allows the brain to gauge the muscle's tension and elongation.

POSITIONAL SENSE

Muscles contain many tiny sensors, known as neuromuscular spindles. These are modified muscle fibres with a spindle-shaped sheath or capsule and several types of nerve supply. The sensory or afferent nerve fibres, which are wrapped around the modified muscle fibres, relay information to the spinal cord and brain about muscle length and tension as the muscle stretches. Signals are then sent back through motor neurons to the muscle to tell it to contract, thus restoring muscle tension to normal. Similar receptors are found in ligaments and tendons. Together they provide the body's innate sense of its own position and posture, called proprioception.